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Resnet-53 for Alzheimer's Disease Detection from MRI Images and Analysis with SVM Tuning with Hyper Optimization Technique

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ABSTRACT—The research utilizes a dataset containing biomarkers and cognitive traits, utilizing advanced machine learning algorithms for early detection and accurate diagnosis of Alzheimer's disease globally. Gaussian filter and bilateral filter to produce an adaptive filter, combining it with Gaussian filtering allows the model to optimize the smoothing and enhancement process depending on local characteristics by adaptively filtering various sections of the picture. After reducing the noise from the images, this output is given to the RESNET for extraction of features from the images. Here, the customized three layers of RESENT-50 to RESNET-53. As a result, feature learning may become more efficient and discriminative. Brain biomarkers can be used to classify patients using information from multi-modal imaging, such as DTI and structural MRI. CNN techniques have been shown to be effective tools for enhancing image-based classification recently. In order to reduce residual errors, the process begins with the extraction of features using ResNet, with a preference for skipping connections. ResNet50 was selected because to its outstanding picture analysis and classification skills. The model's parameters are adjusted in accordance with the difference between the predicted and actual class scores. The SVM model is fine-tuned in the final layer, particularly with regard to Alzheimer's detection for binary and multiclass assignments. In order to improve the model's performance on the validation set, Bayesian optimization using Hyperopt methodically investigates the hyperparameter space by improving parameters like regularization and kernel selection.
In this proposed work, we have used transfer learning with Resnet-Based Image Analysis with SVM Tuning to conduct different classification of Alzheimer disease. These kinds of representations are learnable from the data using deep learning algorithms. The SVM model is fine-tuned in the final layer, particularly with regard to Alzheimer's disease detection for multiple lables. To improve the model's performance the validation set, Bayesian optimization using Hyperopt methodically investigates the hyperparameter space by improving parameters like regularization and kernel selection. Including the use of multimodal data improves the model's durability and offers a thorough and dependable early detection tool.

Keywords— Neuro-Imaging, Gaussian Filter, Bilateral Filter, Adaptive Filter, Hyperopt, regularization, , Hyperplane, Bayesian Optimization CNN.

I. INTRODUCTION

Many neurological diseases, such as dementia, affect a significant percentage of people worldwide. [11] After the age of 60, Alzheimer's patients begin to exhibit increasingly pronounced symptoms. However, in a few instances, the symptoms may appear early (30–50 years old) due to specific gene anomalies. The recommended approach facilitates timely therapeutic therapy and helps in early detection and prevention of AD, according to Rizwan Khan et al. [20]. 315 T1-weighted MRI pictures that were taken from the ADNI database made up the dataset that was used in the investigation. The MRI scans' 3D voxel data is subjected to GM extraction; the resultant GM slices are then utilized to train VGG architectures. Using pre-trained VGG 16 & VGG 19, which were trained on the ImageNet database, a layer-wise transfer learning strategy is implemented, incorporating step-wise block freezing. Tools like the SPM12 are used for operations including segmentation, normalization, registration, and skull stripping. The designs VGG 16 and VGG 19 are used since their performance in feature extraction and image processing applications has been demonstrated. Leveraging the weights of the pre-trained.

Methodologies from the ImageNet database is achieved by transfer learning. [18] For multi-class classification, fresh fully connected layers are integrated with the frozen convolutional base using the softmax function and categorical cross-entropy loss. Threshold-based segmentation methods require choosing appropriate threshold values. Selecting the correct thresholds can be subjective and may require domain expertise. Image processing techniques may not account for the dynamic nature of the disease over time. Below section discusses the popular filters for performing segmentation on images.

II. EASE OF USE

A. Bilateral filter

Combining weights from the spatial kernel and the range kernel to calculate the final weight for each pixel in the neighbourhood. Pixels with similar intensity or colour values are given higher weights. [3] The weights for each pixel are established by combining the range and spatial kernels. It specifies a range kernel that weighs pixels

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according to how similar their values are. Pixels with comparable color or intensity values are assigned larger weights. Several operations are carried out in this, including spatial, range, blending weights, and filtering procedures. The merits are edge preservation, non-linear filtering, customizable parameters, and applications. In this, some operations are performed like spatial, range, mixing weights, and filtering processes. [19] The merits are edge preservation, non-linear filtering, customizable parameters, and applications.

B. Gaussian Filter

In the kernel, the GF is discretized into a 2D matrix known as the GK. In the convolution, for each pixel perform a convolution operation, which involves multiplying the pixel values in the kernel's neighbourhood by corresponding values in Gaussian kernel and summing up the results[19]. GF is utilized mostly for noise reduction, edge preservation, & pre-processing. Gaussian filters do not explicitly preserve edges in the image. While they tend to smooth transitions between regions of different intensity, they can also blur important edges and boundaries. This may impact the detection of structural abnormalities associated with Alzheimer's disease [13].

Grid search is used to optimize hyperparameters, which is crucial to machine learning model optimization. a grid with different test locations and hyperparameter values for every dimension. The process evaluates and trains the machine learning model for every combination of grid hyperparameters. a thorough exploration of the combinations in the hyperparameter grid. This comprehensive method excludes no configuration and investigates the hyperparameter space. The evaluation measure is used to record the model's performance. Grid search finds the best combination of hyperparameters. Although grid search is thorough, it can be computationally expensive, especially for large hyperparameter spaces. In order to balance exploration and efficiency, practitioners mix grid granularity with computer resources and temporal constraints.

C. Two different optimization techniques, each with unique properties and methods, are utilized in machine learning for hyperparameter tuning: hyperopt and grid search. Grid search is a simple yet thorough technique that iteratively investigates predetermined hyperparameter values over a grid, assessing every combination on its own. Despite being simple to use, grid search can be computationally intensive since it is not flexible throughout the optimization process, especially in high-dimensional hyperparameter spaces. Hyperopt model involves the steps as it defines objective function, to take hyperparameter as input trains and define search space to specify the range and type of the parameter. Then hyperopt run the optimization by using ‘fmin’ to perform. And hyperopt uses Tree of Parzen Estimators (TPE) algorithm.

Optimization Techniques: The primary factor for neural networks' remarkable success in deep learning is their capacity to gather complex representations from data, which enables them to take on a variety of tasks that were previously thought to be beyond the scope of conventional machine learning techniques. The development of excellent optimization methods which encourage the efficient training of deep neural networks is one of the primary elements leading to this accomplishment. Numerous optimization strategies have been investigated in this area by academics, such as grid search, sequential approaches, random search, and Bayesian optimization using Hyperopt. Additionally, Bayesian optimization techniques, exemplified by the Hyperopt library, have gained traction in microgrid optimization, as they can efficiently navigate complex parameter spaces by building a probabilistic model of the objective function and using it to.

III. LITERATURE SURVEY

Karim Aderghal et al [1] The newly developed approach focuses on a tiny ROI, such as the hippocampus area, employing CNN for automated categorization of brain images. LeNet-like network architecture is employed, and models are constructed and fused for classifying AD. The technique examines several transfer learning techniques, including sMRI, DTI, cross-domain methodology utilized in MNIST data set, and (iii) a hybrid methodology incorporating two kinds. The suggested technique, which makes use of glib CNN, is appropriate for the less resolving in both techniques. Even with tiny datasets, which are typical in medical image analysis, it still produces meaningful findings. Cross-modal transfer learning, & a combination of the two utilising a shallow LeNet network are some of the transfer learning strategies that are compared. [15] The method avoids 3D conv & full-brain utilisation by using the "2-D+" methodology previously devised for the hippocampus area. According to the interpretation of the data, filters taught on one modality have similar geometrical properties and only need modest adjustments when applied to another modality. The outcomes show that good performance may be achieved even with tiny datasets and few ROI slices. When compared to existing methods, the proposed methodology achieves good accuracy scores using a shallow convolutional network.

Atif Mehmood et al [2] focuses on the early detection of MCI using MRI in the treatment of dementia, particularly AD. Deep learning architecture, The lack of labelled datasets for the model's training is addressed by using particular techniques, such as layer-wise transfer for learning and tissue segmentation. The study draws on information collected in the ADNI database, which includes 85 NC clients, 70 EMCI patients, 70 LMCI patients, & 75 AD patients. To isolate GM tissue from brain pictures and help in diagnosis, tissue segmentation is done. Analysis of comparative research shows that the suggested model performs better when it comes of testing accuracy than current state-of-the-art models.

Liuqing Yang et al [3] DL methodologies in distinguishing related patients effected with AD, MCI, or no signs of dementia were seen utilising base brain MRI results. From structural MRI scans, DL algorithms extract characteristics that are then coupled to additional biomarkers to form an AD prognostic hallmark. The AD predictive signature offers enrichment options in AD clinical trial planning and assists in comprehending patient variability within a research cohort. Enrichment of clinical trials with patients showing fast disease progression is crucial for successful drug design in AD. The article

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demonstrates the use of DL-derived AD features combined a prognostic profile that accurately predicts the transition between MCI to AD after thirty-six months by combining it with other biomarkers.

Utkarsh Sarawgi et al [4] the presented architecture utilizes three different features in a multimodal ensemble approach to identify & grade AD based on MMSE scores. Initially the test was utilizing the ADReSS competition data set, related to the proper datasets with no biases against any one subject and is accessible through DementiaBank. The work highlights the effectiveness of using spontaneous speech as a source for robust inductive transfer learning models, showcasing generalizability through a task-independent feature space. To support deductive methods for AD classified & validation scores, the architecture makes use of domain knowledge. For MMSE score regression and AD classification, the system uses specialised ANN having time properties. The provided method illustrates the value of utilising domain expertise and deductive transference for AD diagnosis and severity evaluation, as well as the possibility of multimodal techniques.

Yu-Ching NI et al [5] focuses on using Tc-99m-ECD SPECT brain perfusion pictures in normal clinics for objective evaluation of AD. The Inception v3 network framework is used in two-stage transfer learning, including pre-training using the ImageNet data set and ADNI database. Three-dimensional images are reorganized into ensemble learning and a two-dimensional sets for information augmentation are used to increase training precision. It is examined how well initial training variables for Tc-99m-ECD SPECT pictures can differentiate between AD and NC. The impact append on the little size in pre-training data from F-18-FDG PET images on model performance is analyzed. A deep learning algorithm utilising SPECT ECD perfusion pictures has been suggested, with pre-training on PET FDG metabolic imaging, shows increased sensitivity and accuracy in differentiating AD from NC.

Amir Ebrahimi et al [6] emphasises applying the ResNet-18 Methodology on MRI to identify Alzheimer's disease (AD). For AD identification in earlier investigations, 2D CNNs were applied to 2D image slices of 3D MRI images. To transfer information from 2D to 3D datasets, the proposed technique uses transfer learning in 3D CNNs. The dataset was derived from the ADNI contains MRI scans. Image pre-processing techniques such as intensity normalization and registration were applied to the MRI scans. Due to the little dataset, data augmentation is utilized for enhancing the classified performances. Transfer learning from ImageNet datasets, pre-training a ResNet-18 Methodology, was recommended to avoid overfitting and leverage knowledge from natural images. The fully-connected layer of the ResNet-18 Methodology was adjusted for AD detection, with a higher learning rate for weight and bias. To adapt ResNet-18 for 3D MRI scans, 2D filters were extended to 3D filters, and the remaining layers were adjusted accordingly.

TABLE I. MERITS AND DEMERIT ANALYSIS ON DIFFERENT APPROACHES

Author	Algorithm	Merits	Demerits	Accuracy
Utkarsh Sarawgi et al	Ensemble methods	It can effectively identify spontaneous speech.	By removing unwanted words, the efficiency can be increased. Text analysis was not performed and the error rate is high. When large images are inputted, the time complexity increases.	88.0%
Yu-Ching NI et al	CNN	Augmented CNN has used limited phenotypes, which have achieved good performance.	Pre-processing can be done for more efficiency.	86%
Amir Ebrahimi et al	ResNet 18	Easy conversion of 2d to 3d filters using proposed method. The images are detected from all possible ways.	No use in increasing resnet layers. Because 2d images are used for only tensor.	96.88%
Heta Acharya et al	AlexNet	Compared with previous method proposed method has 12% increase in performance.	When large images are inputted, the time complexity increases. The issues with computer associates are not solved.	95.70%
Manu Raju et al	VGG16	The performance was high and classify multiple classes.	It can classify only images related data.	99%
Rizwan Khan et al	VGG	It can efficiently identify the new features and achieved good performance..	Both analysis and performances has to be improved and Proper comparison is not defined.	97.89%
Yi Li et al	CNN, transfer learning.	By evaluating separately, the data has acquired high performance.	Target labels are not identified accurately. The test has been performed in the final stage, which may decrease the accuracy.	91%

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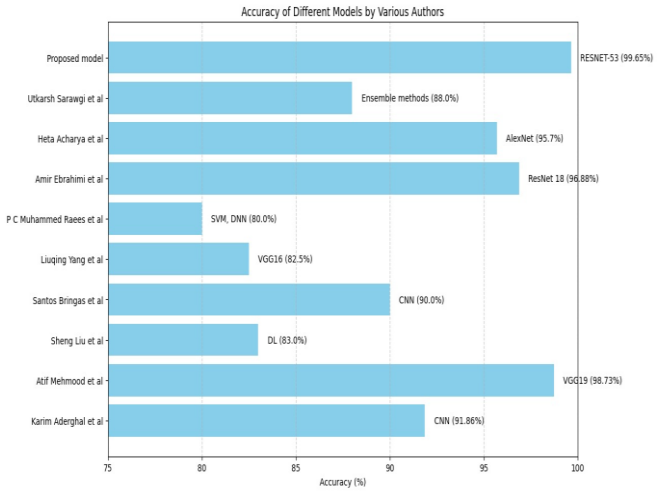

Fig 1. Accuracy Comparison among different Authors with different models.

IV. PROPOSED METHODOLOGY

Recommended approach customizes pre-trained model "ResNet50V2" to extract illness features. The model that was suggested does not feed the input images straight to the CNN. Instead, it customizes a few processes, which will be explained below.

Input Pre-processing: It consists of two steps: 1. Zero Centring and 2. Adaptive Filteration.

A. Zero Centring

In the context of Alzheimer's disease image analysis, zero centring can be particularly beneficial as it helps the deep learning model focus on the subtle structural changes and abnormalities within the brain rather than being distracted by global intensity variations. The application of deep learning has revolutionized field of AD diagnosis and assessment, enabling highly effective analysis of medical imaging data.

B. Adaptive Filter Design Using Integration Approach:

Adaptive filtering has become a basic image processing and signal analysis, gaining widespread recognition for its remarkable capability to effectively manage dynamic signal characteristics and mitigate the impact of diverse noise patterns. A novel approach that has shown significant promise in this domain is the integration of Gaussian and bilateral filtering techniques.

The bilateral filter, a non-linear filter, has shown outstanding results in edge preservation and detail improvement. The bilateral filter's unique approach is to not only consider the spatial proximity of adjacent pixels, and but also their intensity similarity when deciding the appropriate filtering response. The Gaussian filter, often used in the field, is well-known for its efficiency in smoothing images and reducing noise, but it frequently fails to preserve sharp edges and fine details within an image. This initial Gaussian filtering helps reduce high-frequency noise and provides some degree of smoothing. After applying the Gaussian filter, follow it with a bilateral filter.

C. Need of frozen layers in pre-trained models

In deep CNN learned features are valuable and can be leveraged for new tasks without starting from scratch. However, to adapt the pre-trained model to a specific task, some layers need to be frozen while others are fine-tuned. A DNN requires a lot of computing and time to train from start. By freezing a substantial portion of the pre-trained model, significantly reduce training time and resource requirements. Freezing layers in a pre-trained model can improve convergence during fine-tuning. Pre-trained models often contain a wealth of knowledge about various aspects of the data. Pre-trained models with frozen layers are particularly useful when we have a small dataset for the target task. In deep learning, freezing layers refers to the practice of preventing the weights and biases of certain layers in a neural network from being updated or fine-tuned during the training process. When we freeze specific layers in the model, i essentially lock their weights and biases in place.

D. Finding optimal weights using transfer learning

TL is an ML paradigm where information gained from one activity is used to another that is unrelated but yet important. When it comes to deep learning, neural networks are typically pre-trained on large datasets for tasks like image classification or natural language understanding. [17]

Global Average pooling: In the field of deep learning, especially in computer vision, global average pooling, or GAP, has gained a lot of traction as a method for retrieving and combining useful information from input feature maps. In order to capture spatial relationships, fully connected layers are frequently used in traditional neural networks. However, by consolidating all of the spatial data into a single value for each feature map, Global Average Pooling (GAP) provides a more effective and efficient method.

Dense Layer: Dense layers, are also called fully connected layers, are a fundamental component of deep neural networks, playing a crucial role in the extraction and transformation of high-level features from input data. At the core of these layers are neurons that are connected to every neuron in the previous layer, enabling them to learn different nonlinear relationships within the data.

Deep convolutional neural network architecture ResNet50 performs very well in tasks involving picture processing and classification. A deep, hierarchical network called ResNet50 guarantees convergence stability and finds a solution to the vanishing gradient issue. The use of transfer learning is made possible by the pre-trained weights it provides on massive datasets like ImageNet. With less labeled input, the network may improve generalization and faster convergence on certain tasks. Because of its latest architecture, it requires less human feature engineering to gather and extract pertinent data automatically. ResNet50 is a popular and effective feature extraction method for image classification.

E. ResNet blocks performs the following

ResNet blocks consists of Convolution Block, ReLu activation function, Gobal Average Pooling, batch normalization and dense layer. Convolution block with 7by 7 layers and consists of 64 filters.

Convolution Block: A 7 by 7 convolutional layer with 64 filters: In order to obtain information from the input

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pictures, this layer performs a convolution procedure using 64 7x7 filters. Each filter looks at the data and input to create 64 different feature maps. Since batch normalization is essential to determine proper initial weights for deep neural networks, it is a crucial component of healthcare imaging tasks. By lessening the effects of weight initialization, batch normalization makes it easier to experiment with different network configurations and training scenarios. Batch normalization is used to normalize the activations after convolution and reduce internal covariate shifts. Maintaining a fixed size and mean for each feature in the mini-batch improves training stability and speeds up resolution.

With all data, it uses normalization technique to get same type of normalization from different values. By using Batch normalization the value is in between -1 to 1 and 0 to 1. By using batch normalizaton, the values are similar. So, the processing time is less. Integrated Covariate shift method is used in batch normalization. To increse 1 parameter the gama function is used as learnable parameter. The learnable parameter is 0.1.

Then the normalized data is given to the Relu Activation function as Max(0,1). Relu activation function is one of the layer in Convolution block of proposed model.

By encouraging sparsity in the network, the ReLU activation function activates a restricted number of neurons in response to input. ReLU makes it possible for ResNet-50v2 to identify complex relationships in the data, which enhances the model's capacity to explain intricate patterns. To efficiently learn and detect the subtle patterns indicative of Alzheimer's disease in medical pictures, the model must be able to capture non-linearities. It facilitates the discovery of subtle patterns and enhances the network's capacity to represent data by substituting zero for the negative values in the feature maps according to the network. Using max pooling, the network may reduce the spatial resolution while still giving priority to the most essential characteristics.

The spatial dimensions of the feature map are downscaled using max pooling to increase translation invariance and lower computing cost. This approach facilitates abstraction and feature retention by selecting the maximum value while retaining the most significant information from each local region. Max pooling functions are as a regularization strategy by eliminating the features that are less informative and reduce dimensionality reduction. There are three fundamental parts that make up the "Bottleneck" or residual block. The primary element of the residual block is the "identity" function, which bypasses the intermediate layers and transfers the gradients directly to the main deep layers.

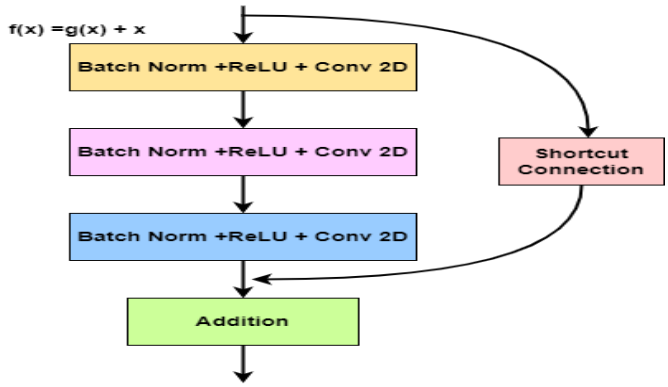

Fig. 2. Working of Residual Block

The bottleneck residual block consist of different convolution layers like 1*1 convolution and 3*3 convolution, here 1*1 and 3*3 represents filter sizes and number of parameters according to the hyper parameters like Regularization (C), kernel(fmin, linear, poly) and gamma. Now we calculate total number of parameters and total multiply accumulate operations.[24]

Identity block is one of the blocks in a specific residual block. It is having three convolutional layers with ReLu activation function and batch normalization.

Hyperparameter Optimization processing steps

- Install hyperopt library.
- Define Hyper parameters with objective function: It takes hyperparameter as input.
- Define search space: Specify range and type of hyperparameters.
- Establish an objective function: To calculate performance of the model with each hyperparameter.
- Using Hyperparameter optimization algorithm from different algorithms i.e., gride search, Bayesian Optimization, Genitic Algorithm and Random search.
- Run hyper optimization
- Train the Final model which is best.

Hyperopt is one of the python library that optimizes hyperparameter optimization and it uses Tree-structured Parzen Estimators (TPE). [21] Hyperparameter tuning is essential for the performance of the machine learning model. Hyperparameter tuning involves hyperparameter space for navigation, manual tuning for default hyperparameter settings and to prevent overfitting.

F. SVM Tuning:

Support Vector Machine is a strong type of algorithm in supervised learning. To increase the performance of the SVM it uses hyperparameter to optimize. [24]

Steps in hyperopt workflow for tuning process is as follows: Defines the hyperparameter with different parameters like regularization parameter or C which manages trade off and kernel type for kernel parameters, Select a search space with the range and type to optimize, Define objective function, apply algorithm, run hyperopt, evaluate and select best for assessment of the performance.

The proposed research uses one versus one computation, defining the objective function as:

$$Objective_Function_SVM = \min\left(\frac{1}{2} * \sum_{i=1}^{n} W_i^2 + C.\sum_{i=1}^{n}\sum_{j=1}^{n} \epsilon_{ij}\right)$$

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where n is the number of classes, W_i are weights and bias terms, C is the regularization parameter, and ∈_ij are slack variables.

The proposed model ResNet 53 initially frozen the first layers and then added 3 layers for the ResNet 50 model.
From layers the features are extracted, so after extraction which feature is important or not we don't know so I have used the first and second steps.
Step 1: First added layer will check whether the extracted features are linear or nonlinear.
Step 2: Then second layer removes the linear data and saves the nonlinear data.
Step 3: The last layer is blocked for fine tuning of SVM by using hyper-parameterization technique to predict the data.

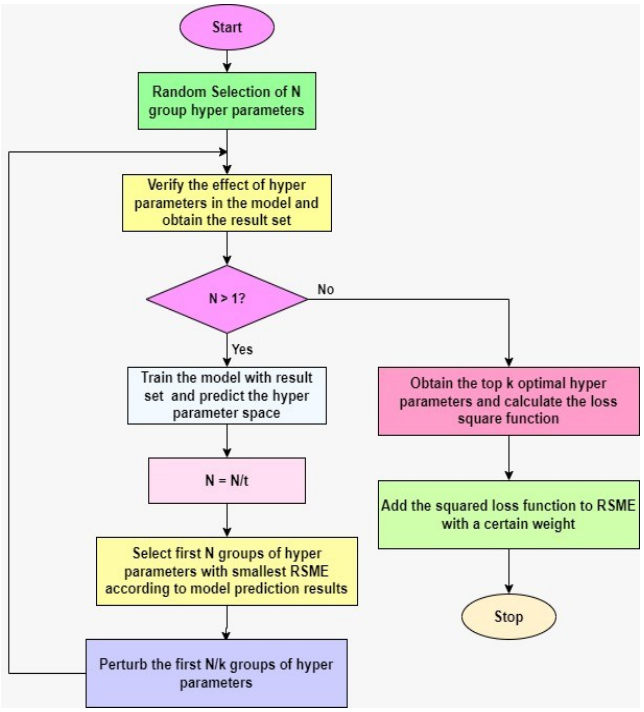

Fig. 3. Hyperopt Work Flow for Tuning Process

[24] The epsilon parameter influences regression tolerance, with lower epsilon causing overfitting and greater flexibility for a vast range of data patterns. By utilizing SVMs for classification and regression tasks, careful parameter altering is necessary, considering the unique features of the dataset. Inadequate fine tuning can lead to overfitting or underfitting, resulting in less than ideal results. The SVM computations are displayed in Figure 4.

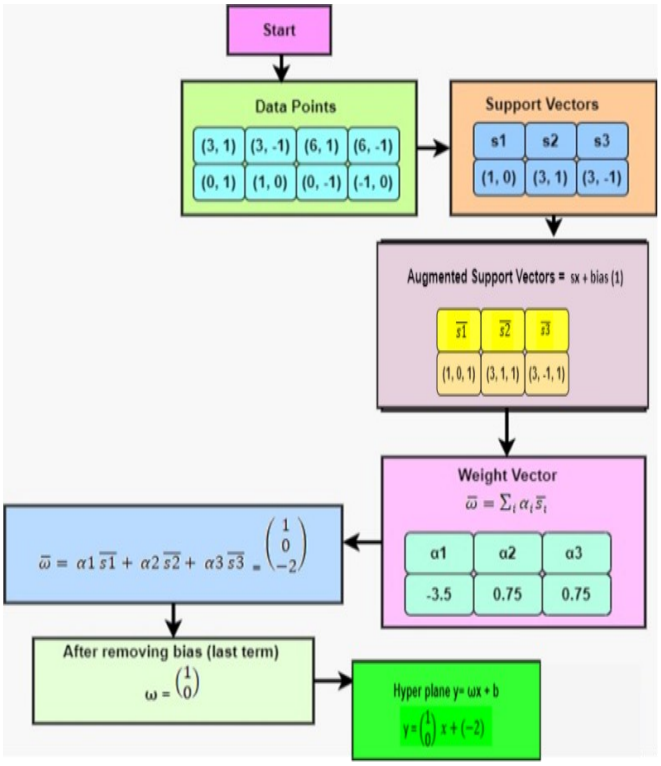

Fig. 4. Kernel Computations in SVM

Hyper Optimization or hyperopt: Here the model finds the probability and likelyhood for each. In each there will be number of probabilities from them max probabilities are picked from each. Then it is used to calculate likelihood, we get output. The proposed model uses kernel, regulation parameter and kernel specific functions to optimize the Hyperparameters.

V. RESULTS & DISCUSSION

Layer (type)	Output Shape	Param #
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 2048)	0
dense_3 (Dense)	(None, 100)	204900
dense_4 (Dense)	(None, 4)	404

Total params: 205304 (801.97 KB)
Trainable params: 205304 (801.97 KB)
Non-trainable params: 0 (0.00 Byte)

Fig. 5. Customized Top Layers of ResNet

Figure 5 shows the top layers of the Resnet architeture and are customised by adding 3 layers of Resnet 50. Parameter training using modified layers in the proposed model, with the GAP layer reducing spatial dimensions and the Dense layer having 2048 units. Global Average pooling is used to extract features from all the images. And Dense layers are used to identift linear or non linear features.

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```
Epoch 38/50
161/161 [==============================] - 1s 8ms/step - loss: 0.0845 - acc: 0.9678 - val_loss: 3.1026 - val_acc: 0.5700
Epoch 39/50
161/161 [==============================] - 1s 8ms/step - loss: 0.0301 - acc: 0.9897 - val_loss: 3.7117 - val_acc: 0.5622
Epoch 40/50
161/161 [==============================] - 2s 10ms/step - loss: 0.0461 - acc: 0.9845 - val_loss: 3.8006 - val_acc: 0.5364
Epoch 41/50
161/161 [==============================] - 2s 11ms/step - loss: 0.0801 - acc: 0.9688 - val_loss: 3.6390 - val_acc: 0.5551
Epoch 42/50
161/161 [==============================] - 2s 9ms/step - loss: 0.1015 - acc: 0.9647 - val_loss: 3.0028 - val_acc: 0.5364
Epoch 43/50
161/161 [==============================] - 1s 7ms/step - loss: 0.0578 - acc: 0.9798 - val_loss: 3.4031 - val_acc: 0.5215
Epoch 44/50
161/161 [==============================] - 1s 7ms/step - loss: 0.0250 - acc: 0.9930 - val_loss: 3.8841 - val_acc: 0.5622
Epoch 45/50
161/161 [==============================] - 1s 8ms/step - loss: 0.0501 - acc: 0.9804 - val_loss: 3.4579 - val_acc: 0.5371
Epoch 46/50
161/161 [==============================] - 1s 8ms/step - loss: 0.0771 - acc: 0.9738 - val_loss: 3.4278 - val_acc: 0.5512
Epoch 47/50
161/161 [==============================] - 1s 7ms/step - loss: 0.0667 - acc: 0.9773 - val_loss: 3.4487 - val_acc: 0.5598
Epoch 48/50
161/161 [==============================] - 1s 7ms/step - loss: 0.1172 - acc: 0.9579 - val_loss: 3.2372 - val_acc: 0.5246
Epoch 49/50
161/161 [==============================] - 1s 8ms/step - loss: 0.0322 - acc: 0.9905 - val_loss: 3.8122 - val_acc: 0.5661
Epoch 50/50
161/161 [==============================] - 1s 8ms/step - loss: 0.0103 - acc: 0.9965 - val_loss: 4.5482 - val_acc: 0.5715
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Fig. 6.Epochs Training using Proposed Methodology

Figure 6 shows a significant improvement in feature extraction accuracy from 57.78% to 99.06% using a customized ResNet-53 model, with minimal training data loss approaching 0.04%.

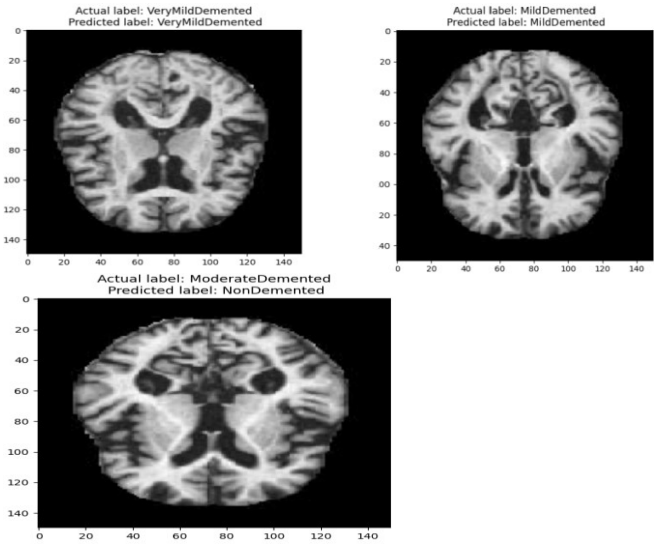

Fig. 7.Classification Reports

If Actual image is moderate and the predicted image is non-demented, then we get chance to get error. We have checked that error case, Figure 7 shows the error case also. The proposed model accurately predicts class labels for diseases like very mild demented, moderate demented, and non-demented, and effectively detects and diagnoses the disease by recognizing its distinctive traits through a comparison process between actual and anticipated class labels.

CONCLUSION

The study focuses on the use of machine learning techniques for the early detection of Alzheimer's disease. It suggests that pre-trained models can be modified to extract specific characteristics from medical images, with additional layers applied to customize the model for Alzheimer's disease detection. These layers should include Global Average Pooling (GAP) and one or more Dense layers, with the number of units in the Dense layers adjusted to regulate feature dimensionality. The research also suggests expanding the use of transfer learning to other neurological disorders or diseases with similar features to Alzheimer's disease. ResNet50 for categorization and visual representation, which compares expected and actual class scores to extract features and modify parameters. Hyperopt and Bayesian optimization are used to fine-tune the SVM model for Alzheimer's disease identification. High sensitivity and specificity were shown by the machine learning model, suggesting that it has the potential to be a trustworthy diagnostic tool. The method's robustness is strengthened by the inclusion of multimodal data, which makes it possible to continuously build precise and accessible diagnostic tools for Alzheimer's disease. The results provide promise for early therapeutic interventions, better patient outcomes, and care. Tested and trained using MRI and PET scans associated with Alzheimer's disease, the suggested ResNet50 model also demonstrates significance in the categorization of medical images.

FUTURE WORK

Subsequent investigations will examine the efficacy of pre-trained ResNet50 classification models on other datasets, including as real-time MRI and PET data, as well as OASIS data. T2 weighted MRI and PET scans will be used to give qualitative transfer learning knowledge. In addition, various network models and neuroimaging approaches will be used to study yoga and meditation practices. The classification of Alzheimer's disease (AD) and its possible overlap with other neurological conditions such as multiple sclerosis, Parkinson's disease, dementia, migraines, and epilepsy are the main topics of this study. Neurological problems are investigated using electroencephalogram (EEG) signals and magnetic resonance imaging (MR) images.

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